

Lecture 1: Mechanical Design in the Aerospace Context

UCK 364E Design of Machine Elements

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Fall 2026

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1. What is a machine element? Why is this course different for you?
2. Loads, factors and margins the way the certification codes define them
3. Design allowables: A-, B-, S-basis
4. Stress-strain review: 3-D stress, yield criteria, stress concentration
5. The design process and the three questions every element must answer
6. Selecting aerospace materials: aluminium, titanium, steels, corrosion, temperature
7. The drawing as a contract: tolerances, fits, finish, standard parts
8. Introduction to standards, with bolt and nut standards as the worked example
9. Design principles and the errors that cause accidents
10. Three worked examples

Machine elements on an aircraft

- A **standardised load-carrying component** with established design rules, standards and catalogue data
- Fasteners, rivets, welds, bonds, keys, splines, springs, bearings, shafts, gears
- You will rarely invent one; you will **select, size and integrate** them
- Service failures are almost always failures of an element or a joint, not of a bulk panel

Element	Aircraft location	Governing concern
Bolts, Hi-Loks, rivets	Fittings, lap joints, frames	Fatigue, preload, bearing/bypass
Lugs and fittings	Wing root, gear trunnion, engine mount	Fitting factor, fatigue, fretting
Welds	Steel-tube fuselage, engine mount	Weld fatigue class, NDT
Adhesive bonds	Sandwich, doublers, repairs, spacecraft	Peel, environment, outgassing
Splines, keys, fits	Accessory drives, actuators	Fretting, tolerance stack-up
Springs	Landing gear, valves, latches	Fatigue, surge, relaxation
Rolling bearings	Wheels, hinges, actuators, gearboxes	L_{10} life, spectrum
Gears	Accessory gearbox, rotor transmission	Tooth bending/contact fatigue

Consequence for this course

Fatigue first, joints in the middle, bearings and gears last.

Loads, factors and margins

Europe — EASA

- **CS** = *Certification Specifications*: the airworthiness code an aircraft type must meet
- CS-25 large aeroplanes, CS-23 small aeroplanes, CS-27/29 rotorcraft, CS-E engines, CS-P propellers
- **Book 1**: binding requirements (e.g. CS 25.303);
Book 2: **AMC**, Acceptable Means of Compliance (how to show you meet them)

USA — FAA

- **FAR** = Federal Aviation Regulations (14 CFR); Parts 25, 23, 27/29, 33 mirror the CS numbering paragraph for paragraph
- **AC** = Advisory Circular: FAA guidance, the counterpart of AMC (e.g. AC 25.571-1D on fatigue and damage tolerance)

How to read a reference like “CS-25.625”

CS-25 is the document, §625 is the paragraph: “fitting factors”. The same rule is FAR 25.625. In Türkiye, SHGM adopts the EASA CS texts, so CS is what you will design to.

Free and official — bookmark: CS-25 (Easy Access Rules, CS + AMC): easa.europa.eu → Easy Access Rules for Large Aeroplanes (CS-25)

FAR Part 25 (eCFR, 14 CFR 25): ecfr.gov → Title 14, Part 25

$$\underbrace{\text{applied}}_{\text{loads analysis}} \times \underbrace{\text{factors}}_{\text{code}} \leq \underbrace{\text{allowable}}_{\text{MMPDS}}$$

- **Limit load** (CS-25.301): max. expected in service; no detrimental permanent deformation (25.305)
- **Ultimate load** = limit $\times$ FS, FS = 1.5 (25.303); carry for 3 s without failure
- In aerospace the factor of safety is **prescribed**, not chosen

Applied *in addition* to 1.5 whenever strength is uncertain or may deteriorate:

- **Fitting factor 1.15** (25.625) on any part transferring load between members, its attachments and bearing surfaces
 - not on continuous plating joints, welded joints
- Casting factor 1.25–2.0 (25.621), depending on inspection
- Bearing factor (25.623) for free-fit parts under pounding/vibration
- Fatigue is *not* a factor: handled by 25.571

$$MS = \frac{\text{allowable}}{\text{applied} \times FS \times (\text{special factors})} - 1 \geq 0$$

- Ultimate margin: F_{tu}, F_{su}, F_{bru} with $FS = 1.5$
- Yield margin: F_{ty}, F_{bry} with $FS = 1.0$ at limit load
- Smaller governs; $MS \approx 0$ is efficient, $MS \gg 0$ is weight, $MS < 0$ is a finding

Shigley vs. CS-25

Shigley: $n = S/\sigma$ chosen by designer (1.5–4, Pugsley). Airframe: FS fixed, loads certified, allowables statistical, margin reported.

- **A-basis:** 99% exceed, 95% confidence — single load path
- **B-basis:** 90% exceed, 95% confidence — redundant structure
- **S-basis:** specification minimum, no statistics

Alloy	Form	F_{tu}	F_{ty}	F_{su}	Use
2024-T3	Sheet	434	289	262	Fuselage skin
7075-T6	Sheet	524	462	317	Wing skin, spars
Ti-6Al-4V	Bar	896	827	545	Gear, mounts, fasteners
4340 (1800 HT)	Bar	1793	1517	1076	Landing gear
15-5PH H1025	Bar	1069	1000	655	Fittings, actuators

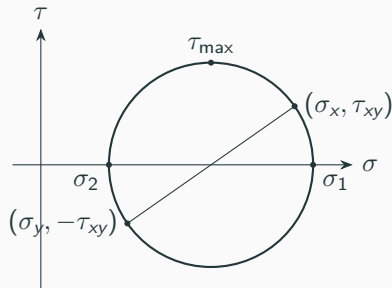
(MPa, approx. B-basis, classroom only)

Stress-strain review

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Third principal stress is zero: the true $\tau_{\max}$ may be $\sigma_1/2$.



Tresca: $\sigma_1 - \sigma_3 \leq S_y/n$

von Mises: $\sigma' = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} \leq S_y/n$

$$\sigma' = \sqrt{\sigma_x^2 + 3\tau_{xy}^2} \quad (\text{axial} + \text{torsion})$$

- Von Mises is the airframe standard; Tresca up to 15% conservative
- Brittle materials (castings, some composites): Coulomb–Mohr — not used here

- $\sigma_{\max} = K_t \sigma_{\text{nom}}$, K_t from Peterson (Pilkey)
- Circular hole, infinite plate, uniaxial: $K_t = 3$; finite width $d/w = 0.2$: $K_t \approx 2.5$
- Static ductile design: usually ignore K_t (local yielding)
- Fatigue: decisive, through K_f

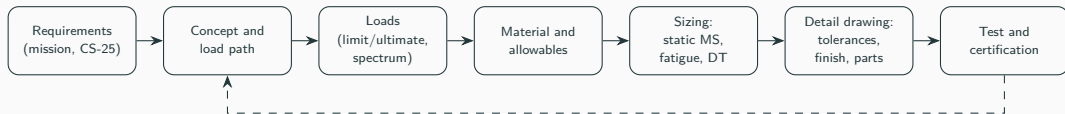
Every lug, fastener hole and window corner is a stress concentration

de Havilland Comet, 1954: square-cornered ADF window cut-outs, pressurisation fatigue.

Case	Stress	Note
Axial	$\sigma = P/A$	net area at holes
Bending	$\sigma = My/I$	$Z = I/c$
Torsion	$\tau = Tr/J, J = \pi d^4/32$	
Transverse shear	$\tau = VQ/(Ib)$	$\frac{3}{2} V/A$ rect., $\frac{4}{3} V/A$ circ.
Bearing	$\sigma_{br} = P/(dt)$	projected area
Thin cylinder	$\sigma_{\theta} = pr/t, \sigma_z = pr/2t$	fuselage
Thin sphere	$\sigma = pr/2t$	bulkhead, tank
Hertz, sphere/plane	$p_{\max} = 3F/(2\pi a^2)$	bearings

Ramberg–Osgood: $\varepsilon = \sigma/E + 0.002(\sigma/F_{ty})^n$, $n \approx 10\text{--}15$ (2024), $20\text{--}25$ (7075)

The design process



- This course lives in boxes 4–6 and reaches into box 7
- 80% of the cost is committed by the time the drawing is released; most errors are born in boxes 2 and 6

1. **Strength**: does it carry ultimate load with positive margin? (static)
2. **Life**: does it survive the load spectrum for the design service goal with the required inspection scheme? (fatigue and damage tolerance)
3. **Function**: does it keep doing its job — no loosening, no fretting, no jamming, no leak, no excessive wear — over the life?

Most failures in service are in categories 2 and 3. Category 1 is the one your textbook spends the most pages on.

Selecting aerospace materials

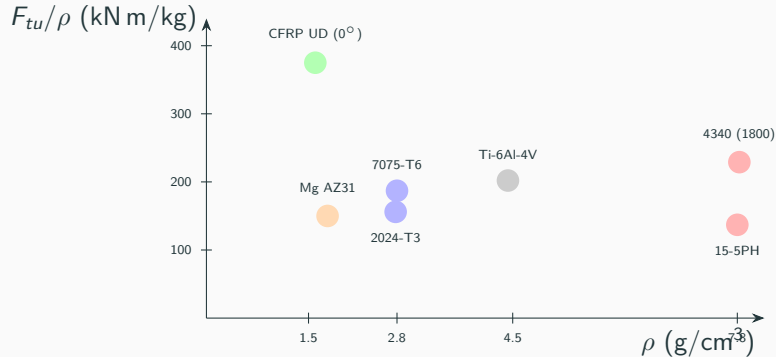
Performance

- Specific strength F_{tu}/ρ , specific stiffness E/ρ
- Fatigue strength and crack-growth rate da/dN
- Fracture toughness K_{Ic} (residual strength)
- Temperature: strength retention, creep
- Corrosion: general, stress-corrosion, galvanic

Practicality

- Availability in the required form (sheet, plate, forging, bar)
- Machinability, formability, weldability
- Certified data: is it in MMPDS?
- Cost, lead time, supplier qualification
- Inspectability (NDT) and repairability

For a *machine element* the questions narrow further: bearing strength, thread strength, wear, and compatibility with the mating part.



Specific strength alone says “use composites and titanium”. Fatigue, toughness, bearing strength, cost and joinability say otherwise for most machine elements.

2xxx (Al–Cu): damage tolerant

- 2024-T3, 2524-T3: fuselage skin, lower wing skin
- Lower strength, higher toughness, slower crack growth
- Poor corrosion resistance — clad or anodised
- Not fusion-weldable in practice

7xxx (Al–Zn–Mg–Cu): high strength

- 7075-T6, 7050-T7451, 7150: upper wing skin, spars, ribs, fittings
- T6 is SCC-prone in the short-transverse direction $\Rightarrow$ T73/T74 overaged tempers for thick fittings

Rules of thumb

- Tension-fatigue dominated $\rightarrow$ 2xxx
- Compression/strength dominated $\rightarrow$ 7xxx
- Thick section, complex fitting $\rightarrow$ 7050/7085 plate or forging, T74
- Weldable structural Al: 6061-T6, 5083 (much weaker), 2219 (launch vehicles), Al–Li 2195/2099

Direction matters

L, LT, ST properties differ; ST toughness of thick 7075-T6 plate can be half of L. Grain direction is a drawing note.

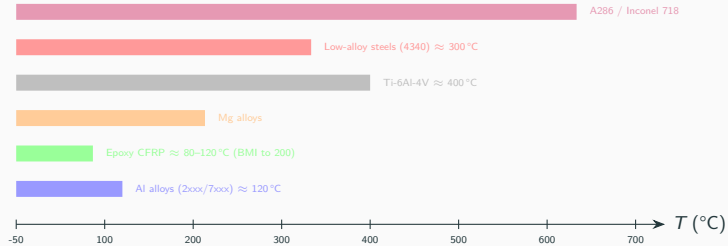
Material	Where	Why / watch out
Ti-6Al-4V	Landing-gear parts, engine mounts, pylon fittings, fasteners, anything touching CFRP	Galvanically compatible with carbon; 400 °C capable; expensive; notch-sensitive; galling in threads (needs lubricant)
Ti-10V-2Fe-3Al, Ti-5553	Landing-gear forgings	Higher strength, replaces steel with weight saving
4340, 300M	Landing-gear main structure, high-load bolts	1800 MPa to 2000 MPa; hydrogen embrittlement from plating; must be shot-peened, cadmium/Zn-Ni plated
15-5PH, 17-4PH, PH13-8Mo	Fittings, actuator housings, pins	Corrosion-resistant, good toughness, easier to machine than 4340
A286, Inconel 718	Hot-section fasteners, exhaust, engine	Strength to 650 °C to 700 °C
Bronze, Al-Ni-bronze	Bushings, sliding surfaces	Bearing pairs with steel pins
CFRP / GFRP	Skins, spars, control surfaces	Element course cares about the <i>joint</i> to it: bearing/bypass, galvanic isolation

- **Galvanic:** Al in contact with CFRP or steel corrodes — use Ti fasteners, glass-fibre isolation ply, sealant, cadmium or Zn–Ni plated steel
- **Stress-corrosion cracking:** 7075-T6 ST direction, high-strength steels; sustained tensile stress + environment
- **Crevice / trap corrosion:** any pocket that holds moisture — drain holes, faying-surface sealant
- **Fretting:** micro-slip at clamped interfaces cuts fatigue life by 3–10×
- **Hydrogen embrittlement:** plated high-strength steel must be baked

Design checklist on every drawing

- Finish and plating spec called out
- Dissimilar-metal isolation specified
- Sealant at faying surfaces
- Drainage path
- Grain direction on 7xxx forgings

Temperature: where each material stops



Brakes, engine bays, anti-ice ducts, supersonic leading edges, re-entry: the element next to the heat source dictates the material and often the whole joint design.

- Load: high, cyclic, with bending; bearing against bushings; must not corrode in wheel-well spray
- Candidates: 4340/300M (highest strength, needs plating and peening), 15-5PH (corrosion-resistant, 1070 MPa), Ti-6Al-4V (light, but galls and has low bearing strength)
- Choice on most aircraft: 300M or 4340, Cd or Zn–Ni plated, shot-peened, running in Al–Ni–bronze bushings — strength and bearing win over corrosion, which is handled by process
- On a small UAV: 15-5PH or 17-4PH — no plating line, corrosion resistant, strength sufficient

Pattern

For *elements* (pins, bolts, gears, springs, bearings) the material is usually a steel; aluminium and titanium dominate the *members* they connect.

The drawing as a contract

A drawing is the legal and technical definition of the part. For an element it must carry, beyond geometry:

- Material spec and temper (AMS/EN number, not “7075”)
- Grain direction for forgings and thick plate
- Dimensional tolerances and **fits** (ISO 286)
- Geometric tolerances (ASME Y14.5 / ISO 1101)
- Surface finish (R_a) where fatigue or sliding matters
- Heat treatment, plating, peening, sealant, primer
- Standard parts by part number (NAS6204-10, MS21042-4)
- Torque or preload for fasteners
- Edge breaks, fillet radii (never “sharp”)
- Inspection requirements (NDT class)

You already read drawings (first year). In this course you will *write* the element-specific notes above.

ISO 286 designation $\varnothing 20 \text{ H7/g6}$

- Letter = position of tolerance zone (H: hole at nominal; g, h, k, n, p: shaft below/at/above)
- Number = grade ($\text{IT7} \approx 21 \mu\text{m}$ at $\varnothing 20$)
- Clearance (H7/g6, H7/f7), transition (H7/k6), interference (H7/p6, H7/s6)

Why you care

- Bearing seats: wrong fit $\rightarrow$ ring creep or preload loss
- Pins in lugs: clearance $\rightarrow$ hammering; interference $\rightarrow$ fatigue benefit

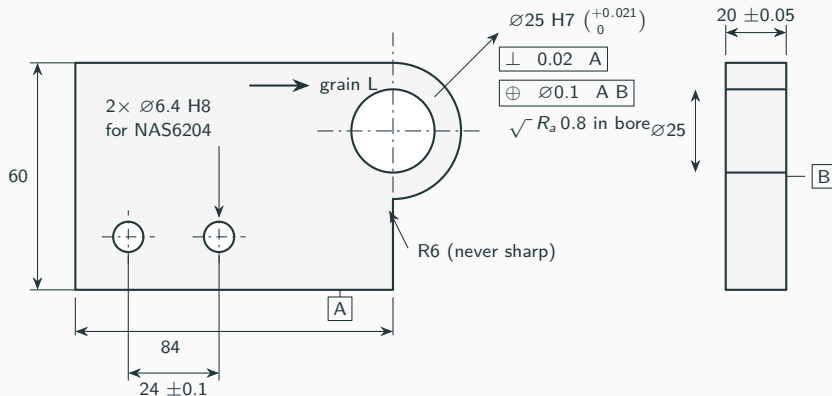
GD&T you will meet

- Position of hole patterns (fastener fit-up)
- Perpendicularity of bearing bores
- Concentricity/runout of shafts
- Flatness of faying surfaces (preload, sealing)

Surface finish

- R_a 0.4 μm to 0.8 μm on fatigue-critical radii
- Machining marks across the load $\rightarrow$ crack starters

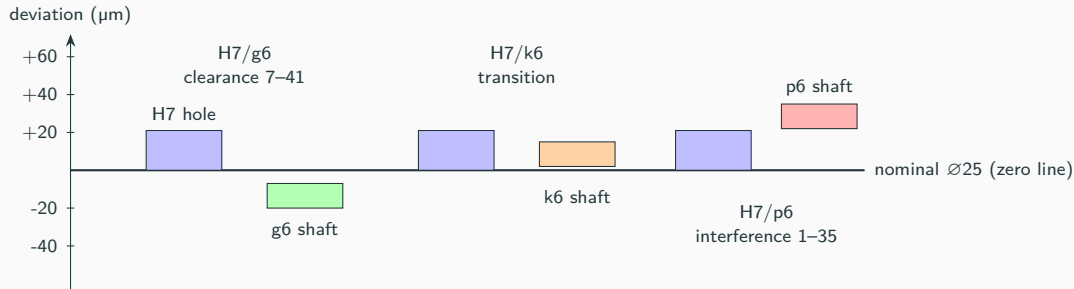
Example: detail drawing of a lug fitting (what the notes do)



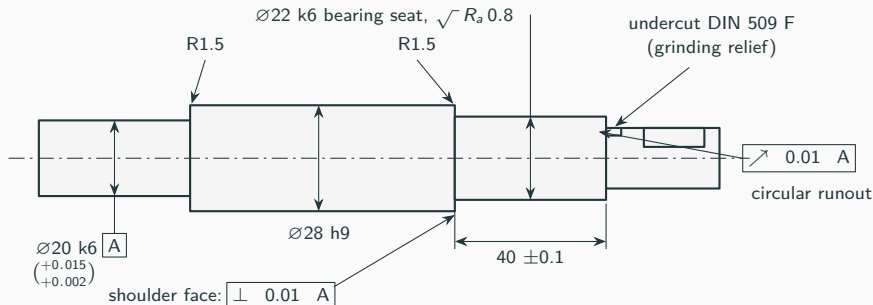
NOTES

1. MATERIAL: 7050-T7451 PLATE PER AMS 4050. GRAIN DIRECTION AS SHOWN.
2. FINISH: ANODIZE PER MIL-A-8625 TYPE II CLASS 1; PRIME PER MIL-PRF-23377.
3. BREAK ALL EDGES 0.3–0.5 UNLESS NOTED. FILLET RADII 2 MIN UNLESS NOTED.
4. BORE Ø25 H7 TO RECEIVE NAS75-16 BUSHING, INTERFERENCE 0.02–0.05.
5. PENETRANT INSPECT PER ASTM E1417, ACCEPTANCE CLASS A.
6. TOLERANCES UNLESS NOTED: ISO 2768-mK. DIMENSIONS IN MM.

Example: tolerance zones and fits (ISO 286) for the same bore



- IT7 at $\varnothing 25 = 21 \mu\text{m}$; IT6 = $13 \mu\text{m}$
- Hole-basis system (H hole, vary the shaft) is the default: reamers and gauges are for standard holes
- Bushing in lug: H7/p6 or s6 — pressed, stays put, pre-stresses the lug (fatigue benefit)
- Pin in bushing: H7/g6 or f7 — rotates, needs lubrication, hammering if clearance grows
- Bearing seats: shaft k6/m6, housing H7/J7 depending on which ring rotates
- A tolerance is a *cost*: IT6 costs 2–3 $\times$ IT9. Tolerance only what the function needs



Everything on this drawing exists because of a machine-element function: k6 so the inner ring does not creep, R_a 0.8 so the seat does not fret, R1.5 fillets smaller than the bearing chamfer so the ring seats against a square shoulder, runout 0.01 so the bearing does not see a wobble, the undercut so the grinder can finish the seat to the shoulder.

Callout	Meaning
NAS6204-10	1/4-28 UNJF close-tolerance shear bolt, 160 ksi, grip 10/16 in
MS21042-4	1/4-28 self-locking reduced hex nut
NAS1097AD4-5	1/8 in 2117 shear-head rivet, reduced head
AN960-416	1/4 in flat washer
MS14101-6	Airframe spherical plain bearing, 3/8 in bore
NAS75-6	Flanged press-fit bushing

- A standard part brings certified strength data with it (NASM1312 tests) — you cite, you don't re-derive
- Mixing metric (ISO) and inch (NAS/MS) systems on one aircraft is a classic source of error; Turkish programmes increasingly use EN/ISO equivalents (EN 6114 etc.)

**Standards: what they are, using
bolts and nuts as the example**

A standard fixes **geometry, material, process, test method and acceptance** so that any supplier's part is interchangeable and its strength is known without testing it yourself.

- **International:** ISO, IEC
- **Regional/national:** EN (CEN), DIN, BS, ASME, ASTM, TSE
- **Industry:** SAE (AS, AMS, ARP), AGMA, AWS
- **Aerospace parts:** NAS (AIA), MS/AN (US military, now NASM under SAE), LN (German aerospace), EN 6xxx/EN 2xxx (European aerospace)
- **Space:** ECSS (Europe), NASA-STD
- **Regulatory:** CS-25, FAR 25 — these are *law*, not standards

The hierarchy you will meet on a bolt

- Thread form: ISO 68 / ASME B1.1 / SAE AS8879
- Dimensions and grip: NAS6204 or EN 6114
- Material and heat treatment: AMS 6415 (4340), AMS 4928 (Ti-6Al-4V)
- Finish: AMS-QQ-P-416 (Cd), AMS 2451 (Zn-Ni)
- Test and qualification: NASM1312, ISO 898-1
- Installation: torque per MIL-HDBK-60 or NASA-STD-5020

ISO metric (ISO 68-1, 261, 262, 965)

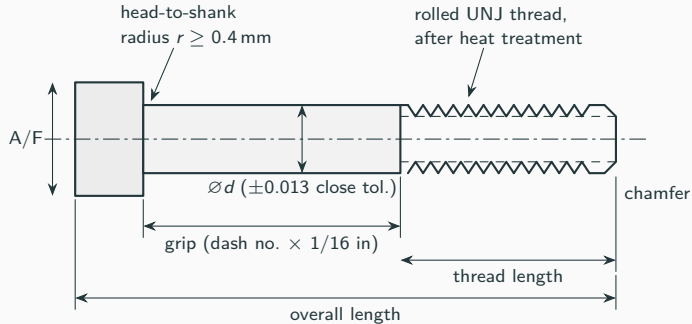
- Designation: M8×1.25-6g (external), M8-6H (internal)
- 60° thread, coarse/fine pitch series
- Tolerance class: number = grade (4–8), letter = position (g, h external; H internal); 6g/6H is default
- Root radius optional $\Rightarrow$ poor fatigue unless “MJ” (ISO 5855, aerospace)

Unified inch (ASME B1.1) and UNJ (SAE AS8879)

- Designation: 1/4-28 UNF-3A (external), -3B (internal)
- UNC coarse, UNF fine, UNEF extra-fine; class 1A/2A/3A loosest to tightest
- **UNJ**: mandatory rounded root radius 0.15–0.18p $\Rightarrow K_t$ from ~ 3.8 down to ~ 2.7 ; fatigue life 2–3 $\times$
- All NAS/MS structural bolts are UNJF-3A; nuts UNJF-3B

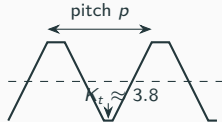
Aerospace rule

Rolled (not cut) threads, rolled *after* heat treatment, with J-root. A hardware-store M8 and an aerospace MJ8 bolt share a nominal size and differ 3 $\times$ in fatigue life.

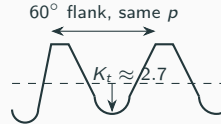


Marking on head: manufacturer, material/strength symbol (e.g. triangle = 160 ksi, 'T' = titanium).

Head-to-shank fillet is the fatigue-critical location on the bolt after the first engaged thread.



UN / ISO M
flat or small-radius root



UNJ (AS8879) / MJ (ISO 5855)
root radius 0.15–0.18 p

- Same major/pitch diameter: a UNJ bolt fits a UN nut, but a UN bolt in a UNJ nut may not (larger minor diameter of the J-nut)
- Rolled thread: fibres follow the root, compressive residual stress $\Rightarrow$ fatigue life 2–3 $\times$ over cut thread
- Tensile stress area $A_t = \frac{\pi}{4} (d - 0.9382p)^2$ (ISO)
- First engaged thread carries $\sim 35\%$ of the load in a rigid nut: the other fatigue-critical spot

ISO 898-1 property class X.Y

- $X \times 100 = F_{tu}$ (MPa), $X \times Y \times 10 = F_{ty}$
- 8.8: 800/640; 10.9: 1040/940; 12.9: 1220/1100
- Nuts: ISO 898-2 class 8, 10, 12 matched to bolt
- Ground machinery, some GA aircraft, ground support equipment

Aerospace strength levels (ksi / MPa)

- 95 ksi (655): low-strength Al or CRES bolts
- 125 ksi (860): AN bolts, 8740 steel — older aircraft
- 160 ksi (1100): NAS6200 series, 4340/8740 — current standard
- 180 ksi (1240): NAS1103–1120 high-strength steel
- 220 ksi (1520): NAS7xxx, H-11 / MP35N — landing gear
- Ti-6Al-4V: 160 ksi, at 55% of the mass of steel

Standards give F_{tu} and shear strength $F_{su} \approx 0.6F_{tu}$ (95 ksi shear for 160 ksi bolts) *as a qualified value*: the allowable comes with the part number, and NASM1312 defines how it was tested (tension, double shear, fatigue, stress durability).

NAS6204-10

- NAS6204: 1/4 in nominal diameter (04 $\times$ 1/16 in), 1/4-28 UNJF-3A
- NAS620x series: close-tolerance shear bolt, 160 ksi, hex head, alloy steel, Cd plated
- -10: grip length 10/16 in = 15.9 mm (unthreaded shank length)
- Suffix letters: D = drilled shank, H = drilled head, T = Ti alloy (NAS6204T10), C = CRES
- Shank diameter tolerance ± 0.013 mm ("close tolerance") for use in reamed holes

EN 6114-4-10 (European metric equivalent)

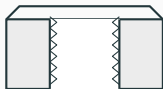
- EN 6114: hexagon-head, close-tolerance, MJ thread, Ti-6Al-4V, 1100 MPa class
- -4: code for MJ6 $\times$ 1 thread, shank 6 mm
- -10: grip code, in 1 mm steps
- Companion nut: EN 6116 or LN 9161 (self-locking, MJ6)

Grip is a design choice

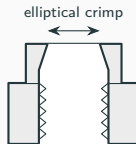
Threads must not be in the shear plane: grip $\geq$ stack-up, then one washer under the nut takes the rest.

Standard	Type	Notes
ISO 4032 / DIN 934	Plain hex nut	Ground machinery only; needs a locking device
ISO 7040 / DIN 985	Nylon-insert lock nut	Not above 120 °C; single use
MS21042	All-metal self-locking, reduced hex, 160 ksi	Standard structural nut; 232 °C
MS21044	Nylon-insert self-locking	121 °C limit; not for engine bays
MS21045	All-metal, 180 ksi	High-strength bolts
MS17825 / AN310	Castellated	With cotter pin; landing gear, control pins
NAS1291	Miniature all-metal lock nut	Ti or CRES; weight-critical
EN 6116, LN 9161	Metric self-locking, MJ thread	European programmes

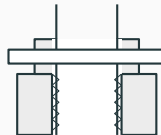
- CS-25.607: no self-locking nut on any bolt subject to rotation in operation unless a non-friction locking device is added; self-locking nuts must meet AS3350 (torque retention after 15 cycles)
- Lockwire (MS33540), safety cable, cotter pin, tab washer — “positive” locking for anything that can back off and hurt someone



ISO 4032 plain
height $\approx 0.8d$



MS21042 all-metal
self-locking, 160 ksi

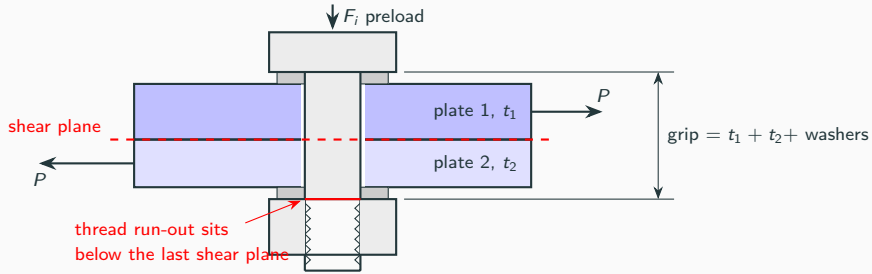


MS17825 castellated
+ cotter pin (MS24665)



MS21044 / DIN 985
nylon insert, $\leq 121^\circ\text{C}$

All-metal self-locking nuts get their prevailing torque from a deformed (out-of-round) top; AS3350 requires the torque to survive 15 install/remove cycles. Castellated + cotter pin is the only positive lock accepted for rotating pins (CS-25.607). Nylon-insert nuts are excluded from hot zones and from primary structure on most programmes.



Choose grip $\geq$ stack-up so the unthreaded shank spans every shear plane; take up the excess with at most two washers under the nut. A thread in the shear plane halves shear area and puts K_t where the load is.

- Washers: AN960 (plain, Cd steel), NAS1149 (metric/inch, various materials), countersunk washers under the head of a bolt with a head-to-shank radius
- Hole: reamed to H7-class for close-tolerance bolts; clearance 0.05–0.1 mm for standard bolts; interference (0.02–0.05 mm) with taper-lok / Hi-Lok for fatigue
- Countersinks per NAS or EN 4xxx, depth $\leq 0.7t$ (knife edge otherwise)

Reading a standard: what to look for

1. Scope — does it cover my size/material/temperature?
2. Dimensions and tolerances table
3. Material, heat treatment, finish references
4. Mechanical properties: tensile, double shear, fatigue, stress durability
5. Qualification and lot acceptance tests
6. Marking — how to verify what is in your hand

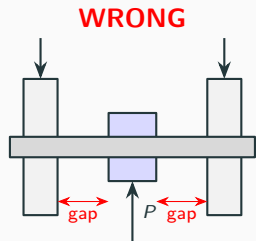
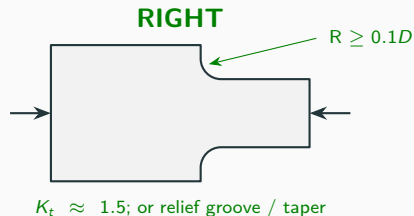
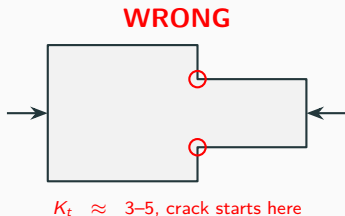
The mechanics (thread stress, preload, torque–tension) and the joint analysis come later. Today: you can read the callout and know what it promises.

Design principles and typical errors

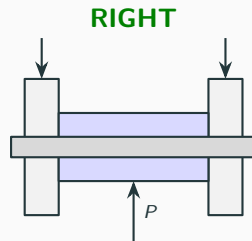
1. **Direct load path**: force goes straight from A to B; every offset is a bending moment
2. **Fasteners in shear, not bending**: a bolt is a pin, not a beam
3. **Stiffness compatibility**: load follows stiffness; a stiff element next to a soft one takes everything
4. **No sharp corners**: radius everything; K_t is where fatigue starts
5. **Symmetry**: avoid eccentric single-shear joints where you can
6. **Self-help**: let the load tighten, not loosen, the joint (preload, taper, direction of thread)
7. **Inspectability and replaceability**: if it cannot be inspected, it must be safe-life with a big factor
8. **Fail-safe or damage-tolerant**: a second load path, or a crack that is found before it is critical

A gallery of common errors (you will see each of these again)

Error	Consequence
Sharp internal corner, no radius on fillet	Fatigue crack at $K_t \approx 3$
Bolt loaded in bending by a misaligned clevis	Bolt fatigue, hole ovalisation
Unspecified preload / no torque note	Joint slips, fretting, bolt fatigue
Thin flange, thick web (stiffness jump)	Load piles into first fastener row
Tolerance stack-up ignored across an assembly	Bearings pre-loaded to death or loose
Weld toe left as-welded at a stressed edge	Weld fatigue class collapses
Dissimilar metals without isolation	Galvanic corrosion, hidden section loss
Pocket that traps water	Crevice corrosion, SCC
Thread run-out in the shear plane	Reduced area, K_t at the worst place
Single load path with no inspection access	Safe-life with no margin for error

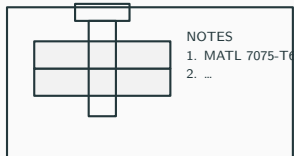


bolt is a beam: $M = Pg/2$,
bolt fatigue, hole ovalisation



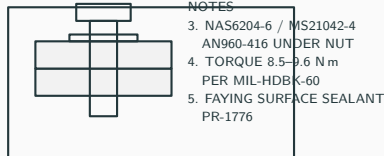
fill the gap (spacer, bushing
flange); bolt in double shear only

WRONG



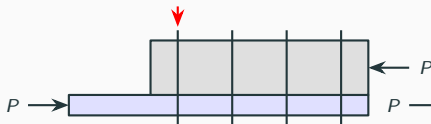
no torque, no washer, no locking called out $\Rightarrow$
shop guesses; joint slips, fretting, bolt fatigue

RIGHT



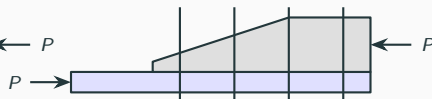
WRONG

60% of P

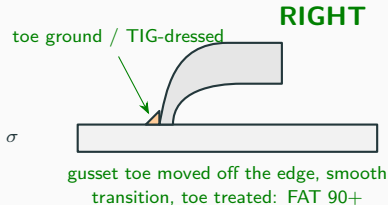
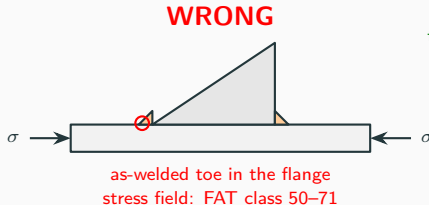
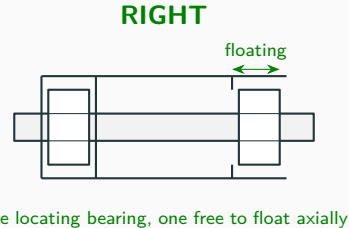
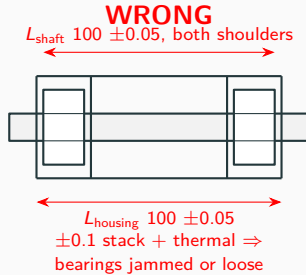


stiff strap, soft sheet: first
row takes most of the load

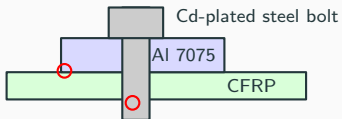
RIGHT



tapered strap (or padding-up the
sheet): load shared 30/25/22/23

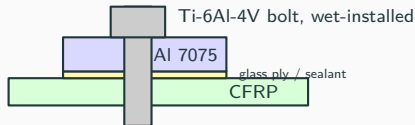


WRONG



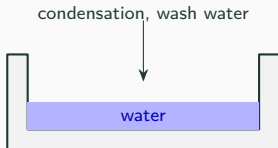
Al-carbon and steel-carbon couples: Al corrodes in months

RIGHT



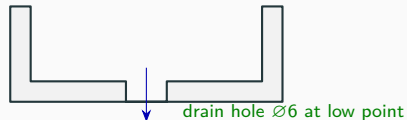
isolate, seal, and pick a compatible fastener (Ti, A286, CRES)

WRONG



closed pocket, lowest point has no exit: crevice corrosion, SCC

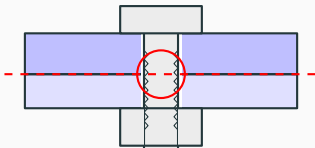
RIGHT



drain, or invert the section, or fill with sealant; primer inside

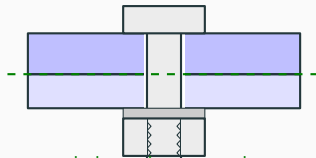
Errors 9–10: thread in the shear plane; single load path with no inspection access

WRONG



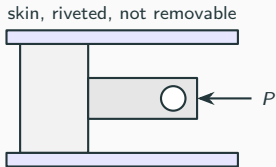
grip too short: thread minor diameter in shear, K_t at the shear plane

RIGHT



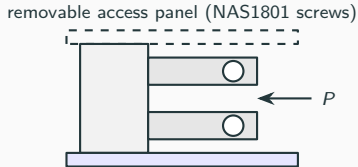
next dash number up, washer under nut: full shank in the shear plane

WRONG



one lug, no second path, no access: crack found only at failure

RIGHT



two lugs each carrying limit alone (fail-safe), visible at inspection

de Havilland Comet, 1954

- Element: pressurised skin cut-out
- Error: square-cornered window, punched rivet holes
- Lesson: K_t and pressurisation fatigue; full-scale fatigue test

Aloha Airlines 243, 1988

- Element: riveted lap joint with cold bond
- Error: bond failure shifted all load to rivets; multi-site cracking
- Lesson: multiple-site damage, inspection intervals

Alaska Airlines 261, 2000

- Element: stabiliser jackscrew and acme nut
- Error: wear without lubrication, no secondary load path
- Lesson: wear as a failure mode, maintenance in the design

None of these was a strength-margin error. All were element, detail, or maintenance errors — the subject of this course.

Worked examples

$$r = 2.0 \text{ m}, \Delta p = 60 \text{ kPa}, 2024\text{-T3 } t = 1.6 \text{ mm}, F_{tu} = 434, F_{ty} = 289 \text{ MPa}$$

$$\sigma_{\theta} = \Delta p r / t = 75 \text{ MPa} \quad (\text{limit}), \quad 1.5\sigma_{\theta} = 112.5 \text{ MPa}$$

$$\text{MS}_{\text{ult}} = 434 / 112.5 - 1 = 2.86, \quad \text{MS}_y = 289 / 75 - 1 = 2.85$$

Lesson

Cabin skin is never sized by static strength. It is sized by crack growth from rivet holes over $\sim 60\,000$ pressurisation cycles and by two-bay-crack residual strength. Static margins are necessary, rarely design-driving.

7075-T7351, $t = 20$ mm, $d = 25$ mm, $P_{\text{lim}} = 180$ kN; $F_{bru} = 1055$, $F_{bry} = 758$ MPa

$$P_u = 1.5 \times 1.15 \times 180 = 310.5 \text{ kN}$$

$$\sigma_{br} = P_u/(dt) = 621 \text{ MPa}, \quad \text{MS}_{\text{ult},br} = 1055/621 - 1 = \mathbf{0.70}$$

$$\sigma_{br,\text{lim}} = 1.15 \times 180\,000/500 = 414 \text{ MPa}, \quad \text{MS}_{y,br} = 758/414 - 1 = \mathbf{0.83}$$

- Without the fitting factor: $\text{MS} = 0.95$ — on a fitting that is a certification finding
- Why does bearing *yield* at limit matter for a pinned joint?
- Net section, shear-out and fatigue come later

$d = 16 \text{ mm}$, $P = 25 \text{ kN}$, $T = 40 \text{ N m}$, 15-5PH $F_{ty} = 1000 \text{ MPa}$

$$\sigma_x = 4P/\pi d^2 = 124.3 \text{ MPa}, \quad \tau = 16T/\pi d^3 = 49.7 \text{ MPa}$$

$$\sigma' = \sqrt{124.3^2 + 3(49.7)^2} = 151.7 \text{ MPa}, \quad \text{MS}_y = 1000/151.7 - 1 = 5.6$$

Statically trivial. Sized later by fatigue at thread run-out ($K_t \approx 3\text{--}4$) and by column buckling.

Next time

- Read CS-25.301–25.307 (Subpart C, loads) and 25.601–25.625 (Subpart D, design and construction), each about one page, in the [EASA Easy Access Rules for CS-25](#); the American twin is [14 CFR Part 25 on the eCFR](#)
- Shigley Ch. 1, 3 (review), 5.1–5.5; Niu Ch. 1–3
- Self-study problems 1–5 in the notes
- Next: stress-life and strain-life fatigue — the heart of the course